

A Statistical Perspective

Another way to measure the exact impact of an untimely investment “famine” on the sustainability of your retirement income is to analyze many sample retirees who experienced good and bad returns at different points in their retirement, and then see who fared better. We don’t have this luxury of data, and it might take a while to see things play out with the baby boomers.

The next best thing to a natural experiment is a diligent research associate with a powerful computer at her disposal. This I have. And together with Anna Abaimova, we generated thousands of possible sample paths for the economic future of a theoretical retirement. We used Monte Carlo techniques—a topic I elaborate upon in Chapter 8, “Spending Your Retirement in Monte Carlo”—to simulate sample paths for inflation, investments returns, health, and longevity. In some of these simulation paths, the hypothetical retiree was “killed” (by the computer algorithm) while still having plenty of money in her retirement account. In other simulation paths, the retiree ran out of money and had to tap other sources of wealth (housing, kids, welfare) to continue spending. The summary results of this analysis are shown in Table 6.2.

TABLE 6.2 Impact of Various “Risk Factors” on Retirement Income Sustainability for a 65-Year-Old (100% Equity Allocation)

Risk Factor	Correlation: Factor and Sustainability
<i>Worse Than Expected Returns During:</i>	
First 7 yrs	–56.3%
Second 7 yrs	–27.5%
Third 7 yrs	–11.0%
Fourth 7 yrs	–2.4%
Fifth 7 yrs	–1.2%
Higher longevity	–53.9%
Higher inflation	–5.8%

Simulation assumptions: neutral spending consumption rate = 9.85%; Equity: expected returns = 11%, volatility = 18%; Inflation: expected inflation = 3%, volatility = 2%
Source: M. Milevsky, “Feast or Famine First?,” *Research Magazine*, December 2007.